

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-14

Faculty

Dr. Rajasekhar N V

Associate Professor G-1

SENSE, VIT-AP University

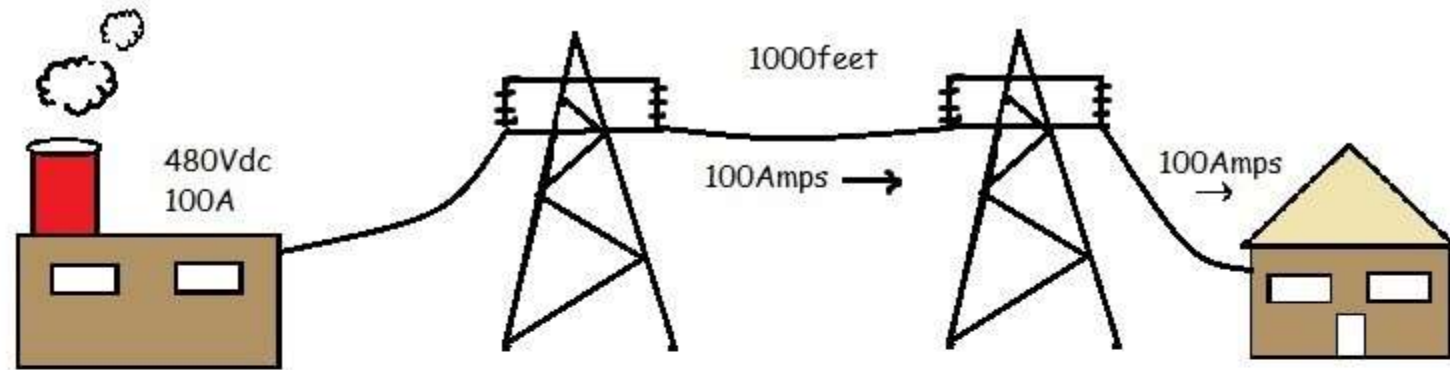
Mail: rajasekhar.venkata@vitap.ac.in

Cabin: Faculty Area 424-C:AB-1

Module-2

AC Circuit Analysis

DC SYSTEM

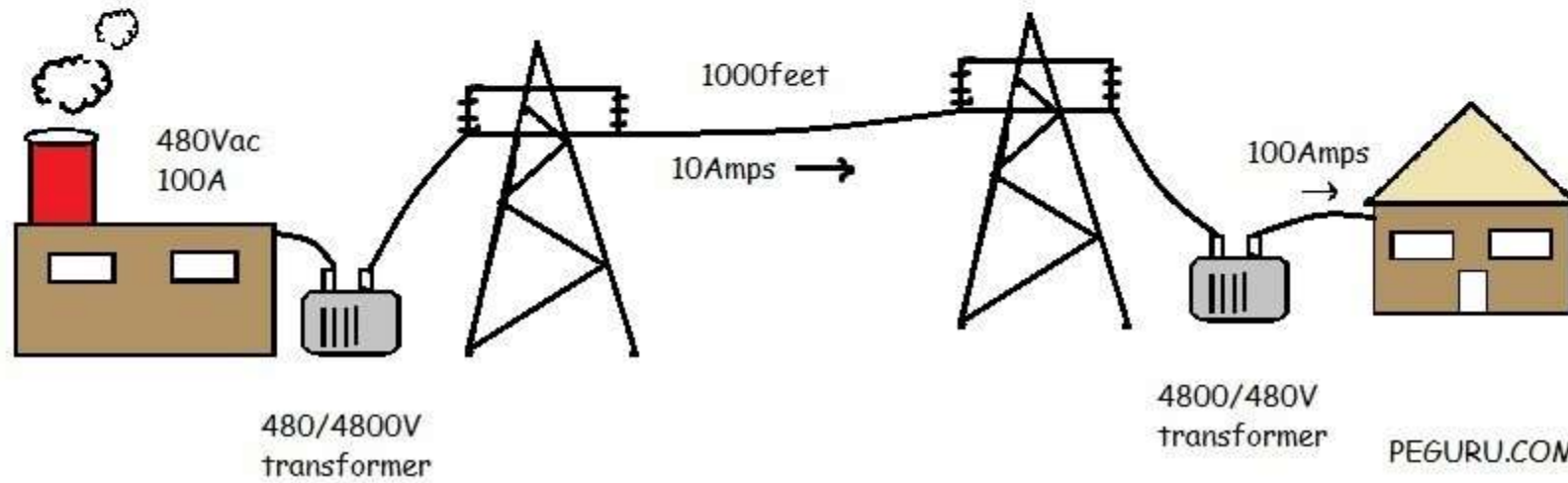


Generating Station

Transmission System

Load

AC SYSTEM



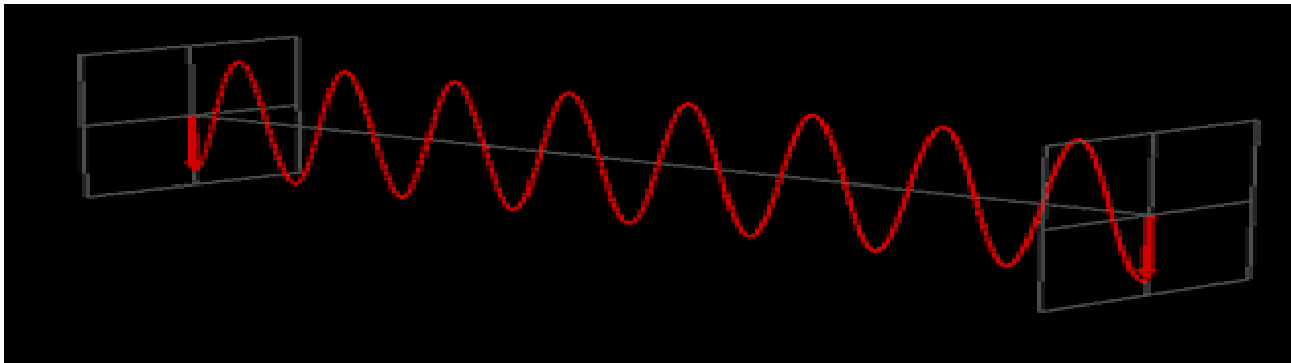
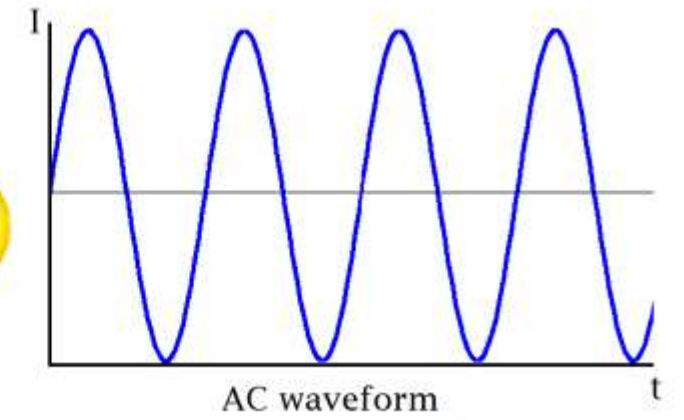
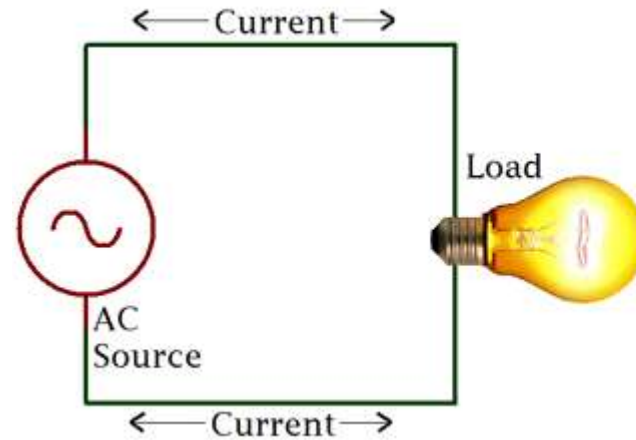
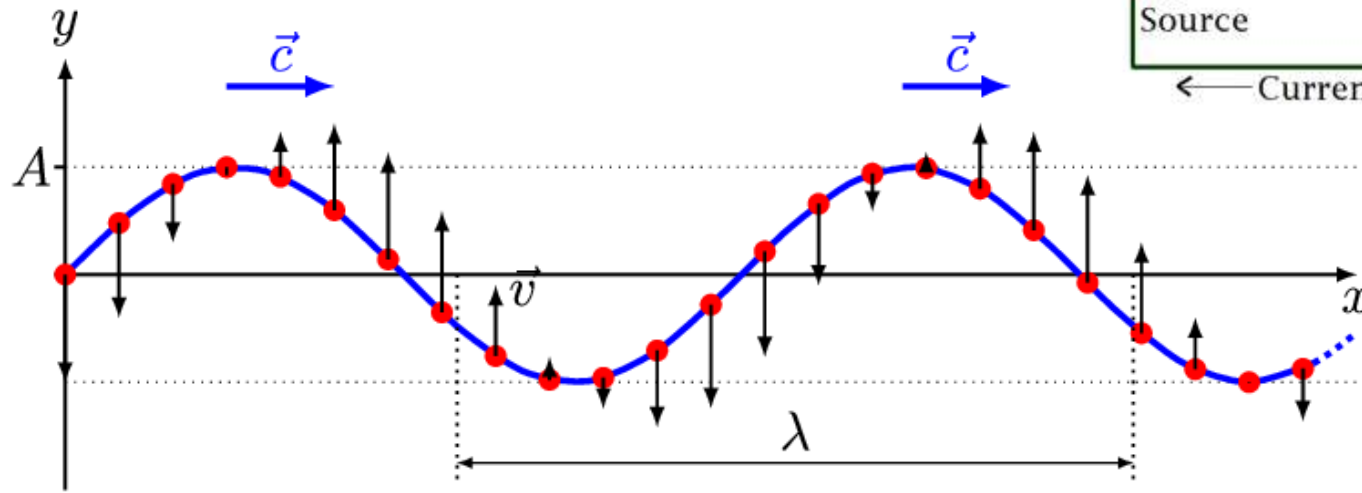
480/4800V
transformer

4800/480V
transformer

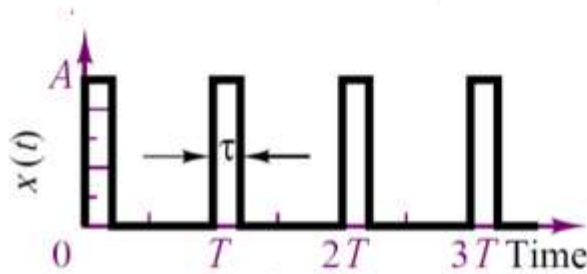
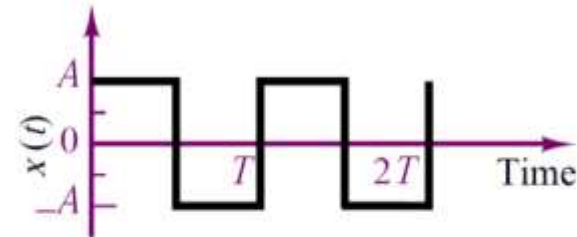
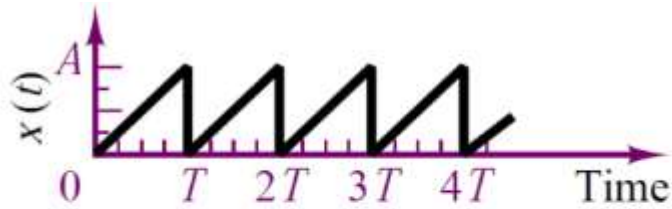
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DC SYSTEM	AC SYSTEM	DC SYSTEM	AC SYSTEM
1. To carry 100Amps over the line, a larger cable (in diameter) will be required for the DC system.	1. After transformation, the current on the power line will be 10Amps. A smaller cable will be required.	4. Losses in transmission system = $VD \times \text{Current}$ (in watts) = $15 \times 100 = 1500$ watts	4. Losses in transmission system (in watts) = $15 \times 10 = 150$ watts. Ten times less than DC transmission.
2. Larger cable means lower conductor resistance. Typically, 0.15 ohms per 1000feet can be used for a 100Amp conductor (per AWG). In which case, Voltage Drop (VD) across the line = $0.15 \times 100 = 15V$.	2. Smaller cable (in diameter) means higher resistance. Typically, a 1.5 ohms per 1000feet can be used for a 10Amp conductor. In which case, Voltage Drop (VD) = $1.5 \times 10 = 15V$. Same as a DC system.	5. Transformers <i>do not</i> operate with DC supply wired to it. It will short-circuit. The only way to step down the voltage for distribution is through a motor-generator set or a rotary converter – an inefficient process.	5. Transformers operate at 99% efficiency at full load. Used throughout the AC system.
3. The DC generator must generate 480V plus 15V to deliver power to the house. At the house, the voltage, therefore, will go from 495V at no-load to 480V at full-load. A 15V variation.	3. Allow taps on the transformer to raise the voltage by 15V to obtain 481.5V. At the house, this is equivalent to 481.5V. A 1.5V variation from no-load to full-load. Engineers call this variation of voltage as the Voltage Regulation (VR). An important factor in the power system. The less the VR the better the system.		

AC Waveform



Some Typical signals



- These signals appear frequently in practical applications and are a useful approximation of many physical phenomena.
- **Cycle and symmetrical waveform**
- **Period and Frequency**

